8 Read the passage below and answer the questions that follow.

To bypass the biological limits of the human eyes in the night, engineers developed the Night Vision Goggles (NVGs) to allow soldiers and rescue workers to see in near-darkness. It converts very faint ambient light into a bright image through photoelectric effect.

Figure 8.1 shows a simplified schematic of an image-intensifier NVGs, comprising a photocathode, a microchannel plate (MCP), a phosphor screen and an eyepiece.

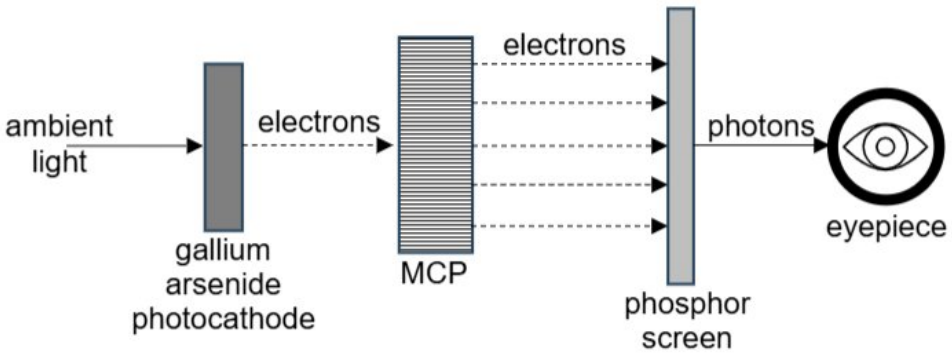


Figure 8.1

Ambient light from moonlight or starlight incidents onto a gallium arsenide (GaAs) photocathode in the NVGs, which has a work function of approximately 1.4 eV. The ejected photoelectrons are then accelerated through a potential difference of 150 V towards the MCP, a glass disc of thickness 480 μm drilled with millions of parallel channels, each of diameter 8.0 μm and at an angle of 13 ° to the normal of the MCP.

When a single electron enters a channel, it strikes the inner wall and releases secondary electrons via secondary emission. Each time an electron strikes the inner wall, 3 secondary electrons are emitted. These secondary electrons are themselves accelerated by a potential difference of 800 V applied across the MCP, striking the wall again and releasing further secondary electrons until a cascade of electrons exits the channel, as shown in Figure 8.2.

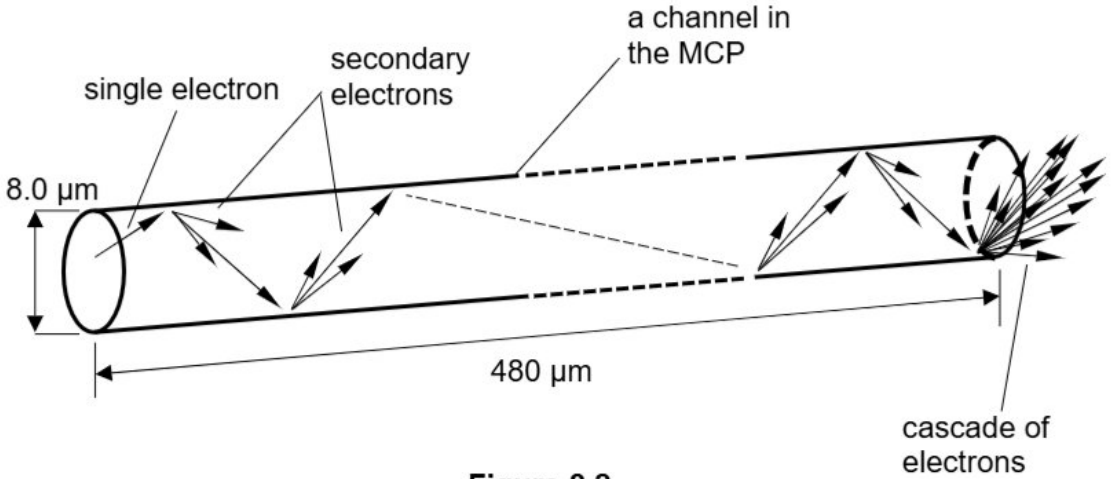


Figure 8.2

The cascade of electrons exit the MCP and is accelerated through a further potential difference of 4500 V across a short distance onto a phosphor screen. These accelerated electrons penetrate deep into the phosphor screen creating electron-hole pairs in the phosphor lattice. From these electron-hole pairs, photons are emitted from the phosphor screen. The emitted light is viewed through an eyepiece lens by the observer.

(a) The work function is defined as the minimum energy required to remove an electron from a metal surface.

(i) Determine the threshold wavelength of gallium arsenide.

threshold wavelength = ... nm [2]

(ii) All objects above absolute zero emit radiation.

The peak wavelength of radiation λ_{peak} is governed by Wien's law:

$$\lambda_{peak} T = b$$

where T is the absolute temperature and b is the Wien's displacement constant which is approximately 2.898×10^{-3} m K.

Explain quantitatively whether the peak wavelength from a human body at a temperature of 37 °C can result in an emission of a photoelectron from gallium arsenide.

.....

..... [2]

(b) Calculate the speed of a photoelectron entering the MCP, assuming that it was emitted from the photocathode with negligible energy.

speed = ... $m\ s^{-1}$ [2]

(c) The number of wall collisions n an electron undergoes in a channel of the MCP is given by

$$n = \frac{L}{2d} \sin \theta$$

where L is the length of the channel, d is the diameter of the channel and θ is the angle the axis the channel makes with the normal to the MCP surface.

(i) Suggest why the channels in the MCP are drilled tilted at an angle to the normal of the disc face.

……

……[1]

(ii) Determine the number of electrons exiting for each electron that enters a channel in the MCP.

number of electrons = …… [2]

(d) Suggest two reasons why the electrons exiting MCP are accelerated through a high potential difference of 4500 V across a short distance onto the phosphor screen.

1. ……

……

……

2. ……

……

……[2]

(e) Each high-energy electron accelerated towards the phosphor screen creates 450 electron-hole pairs in the phosphor lattice, emitting 350 photons of wavelength 545 nm.

(i) Determine the average energy required to create one electron-hole pair.

average energy = …… eV [2]

(ii) The internal quantum efficiency is given by the ratio of the total useful energy emitted as light to the energy of the incident electron.

Calculate the maximum internal quantum efficiency of the phosphor screen.

maximum internal quantum efficiency = ... [2]

(iii) Two spots formed on the phosphor screen subtend an angular separation of 3.3×10^{-4} rad at the observer's eye. In a dark environment, the dark-adapted pupil of the eye has a diameter of 7.0 mm.

Determine whether these two spots can be distinguished by the observer.

.....

..... [2]

(f) A student wanted to determine the value of Planck constant h by radiating gallium arsenide with monochromatic light of varying frequency f and measuring the stopping potential V_s needed to prevent photocurrent.

The equation relating the stopping potential V_s and the frequency of light f is given by

$$eV_s = hf - \phi$$

where e is the elementary charge and ϕ is the work function.

The variation of stopping potential V_s with frequency f obtained by the student is shown in Figure 8.3.

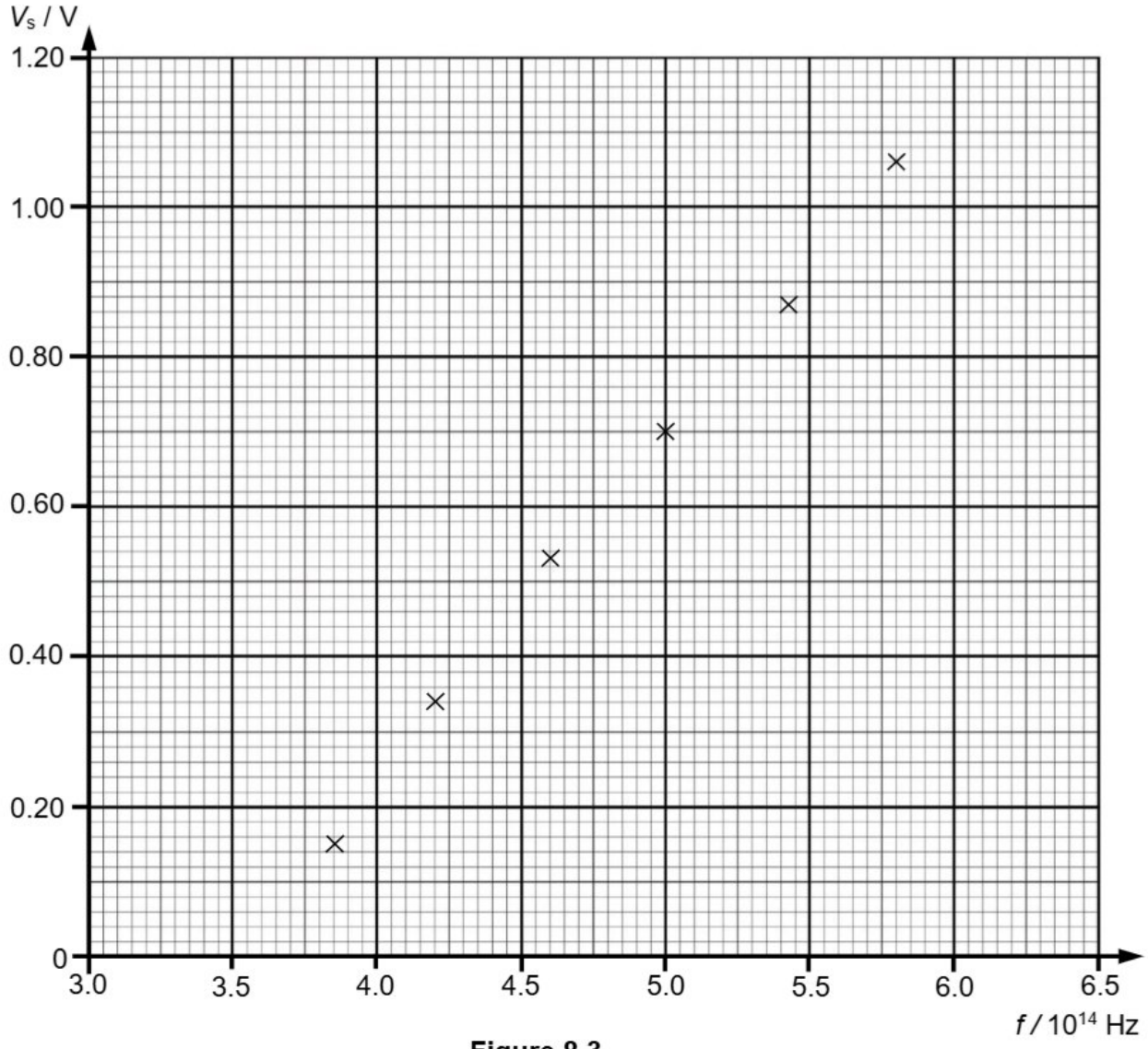


Figure 8.3

Use Figure 8.3 to determine the value of the Planck constant h.

h = ... J s [3]

[Total: 20]